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## ME 204: Materials Science for Engineers

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| Course Code   | ME 204                                                                                          |
| Course Name   | Materials Science for Engineers                                                                 |
| L-T-P-C       | 3-0-0-3                                                                                         |
| Prerequisites | None                                                                                            |
| Intended for  | B.Tech.                                                                                         |
| Distribution  | Core                                                                                            |
| Approval      | 2nd Adhoc Senate (May 2012), in the Minor in Device Materials / Structural Materials curriculum |

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### Course Contents

#### A Structural Applications

- i Static structural applications –stress strain diagram, elastic, yielding and plastic behavior, properties to characterize each, application of metals, ceramics, polymers and composites in static structures like buildings, bridges, furnace structure, etc.
- ii Dynamic structural applications - fatigue, creep-fatigue interaction; Application of materials in automobiles, hydroelectric and thermal power plants.
- iii Manipulation of materials properties through different treatments

#### B Electrical and Electronic Application

Conductors and conductivity, Capacitors, considerations for choice of materials in different applications; metallic and organic semiconductors, p-n junctions, other devices, I-V characteristics, optoelectronic materials and devices, the considerations for the choice of materials; Magnetic materials, Dielectric materials, electrical and magnetic sensors, read- write heads, spintronic devices; superconducting materials and their applications in magnets.

### Text Books

1. V. Raghvan, **Materials Science and Engineering- A first course**, Prentice Hall of India, New Delhi.
2. William D. Callister, Jr, **Materials Science and Engineering- An introduction**, Wiley and Sons, Inc.

### References

1. Kenneth G. Budinski, **Engineering Materials: Properties and Selection**, Prentice Hall, USA.
2. S. O. Kasap, **Principles of Electronic Materials and devices**, 3rd Edition, Tata-McGraw Hill Education Pvt. Ltd.
3. Ben G. Streetman and Sanjay Bannerjee, **Solid State Electronic Devices**, 5th Edition, Pearson-Prentice Hall, USA.